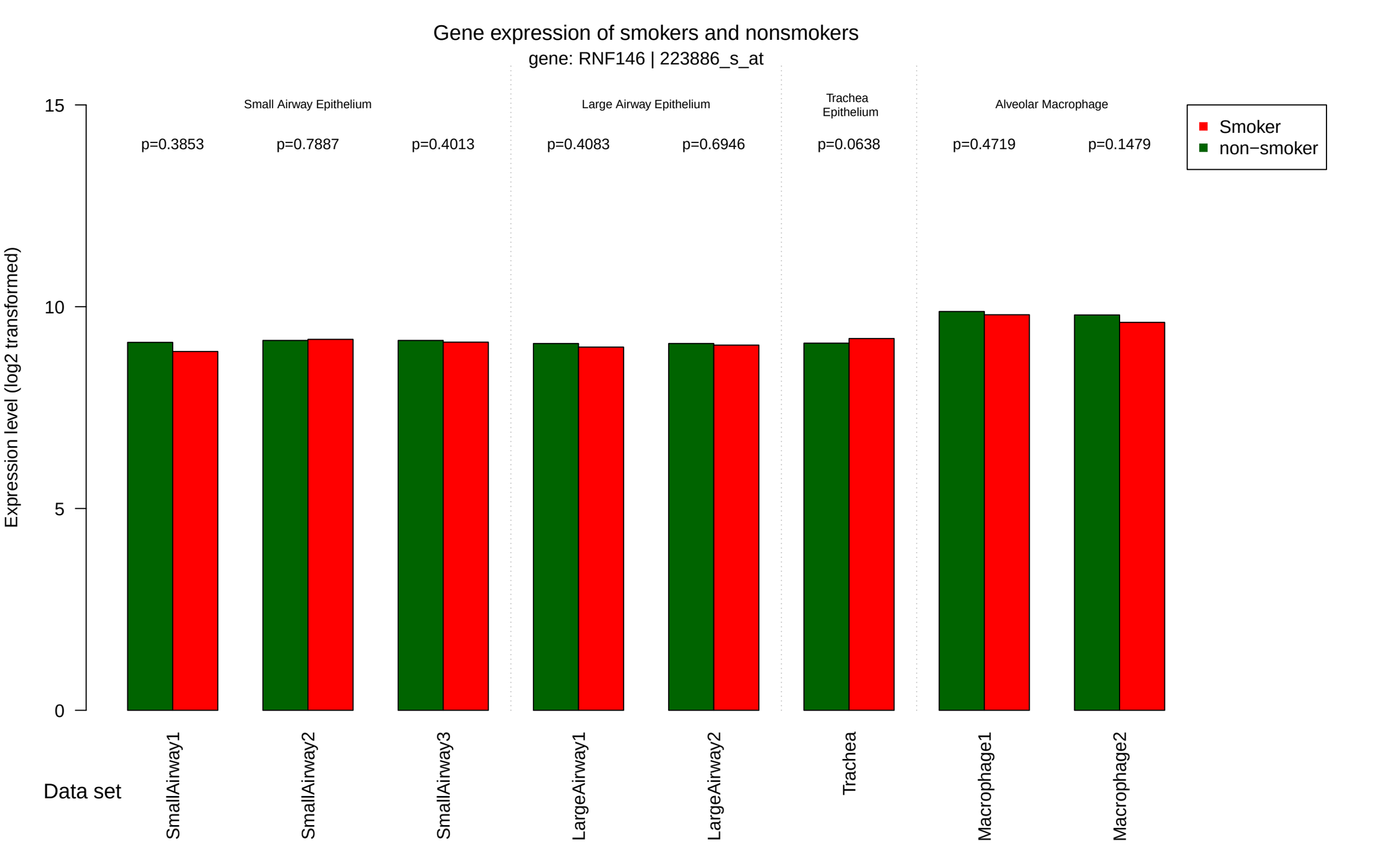
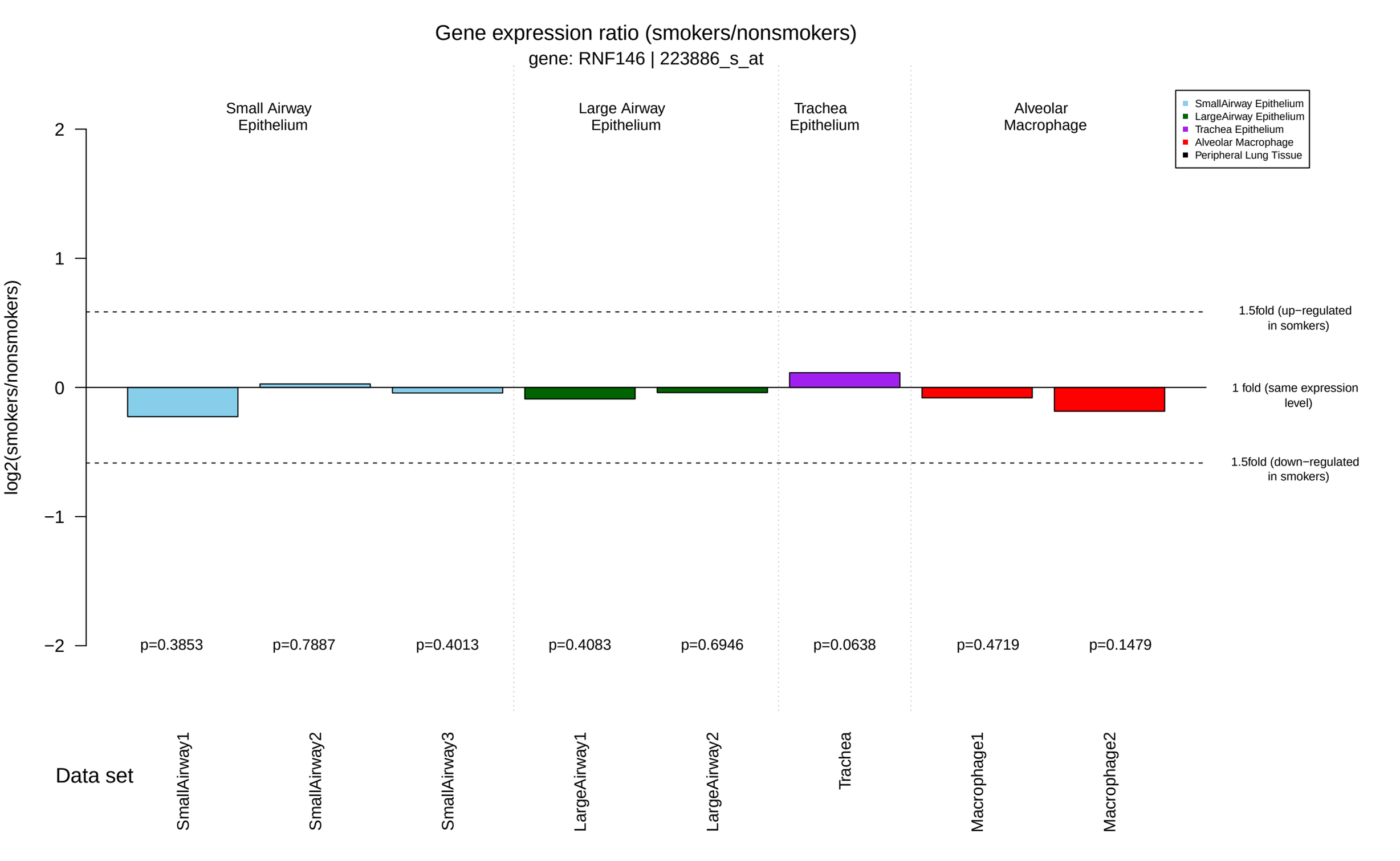
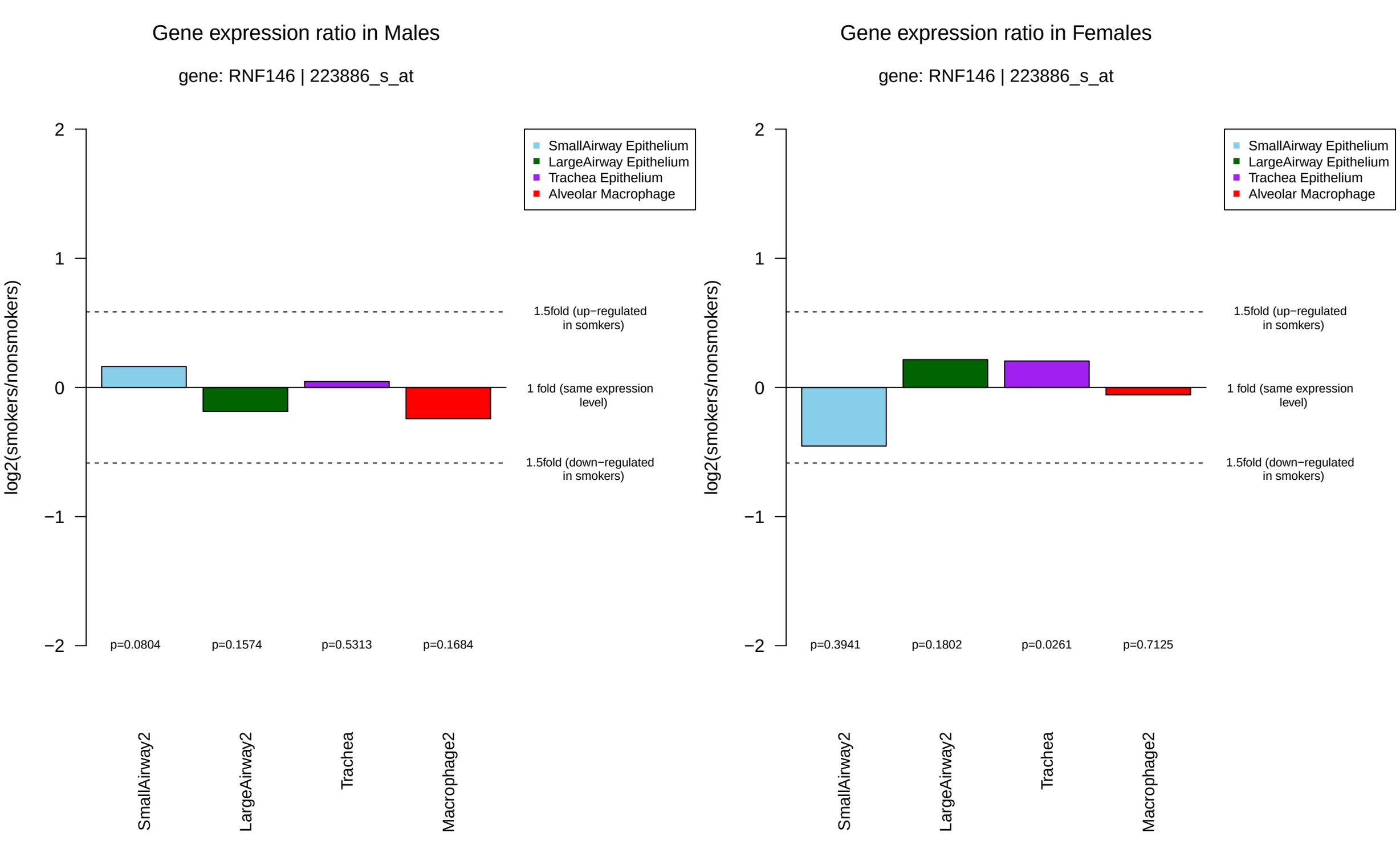
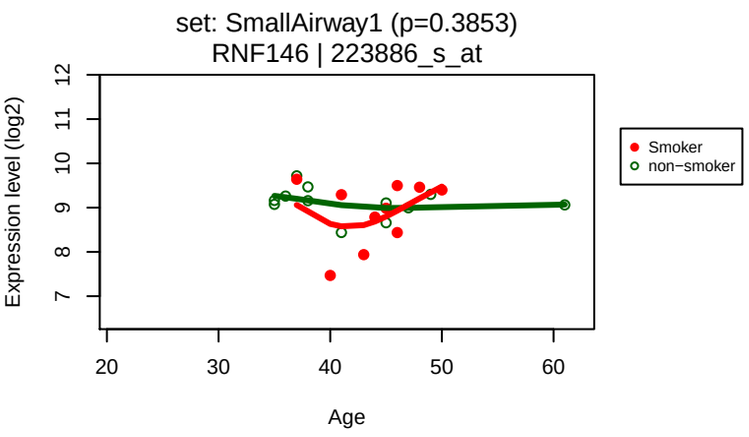
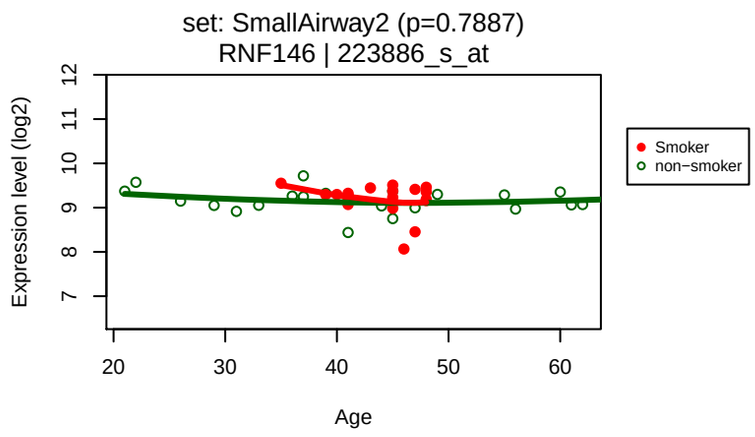
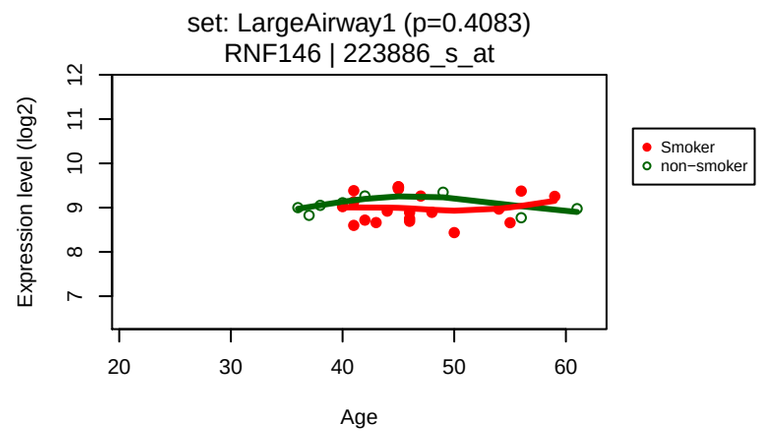

Gene expression of smokers and nonsmokers
gene: RNF146 | 223886_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3853
p=0.7887
p=0.4013
p=0.4083
p=0.6946
p=0.0638
p=0.4719
p=0.1479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

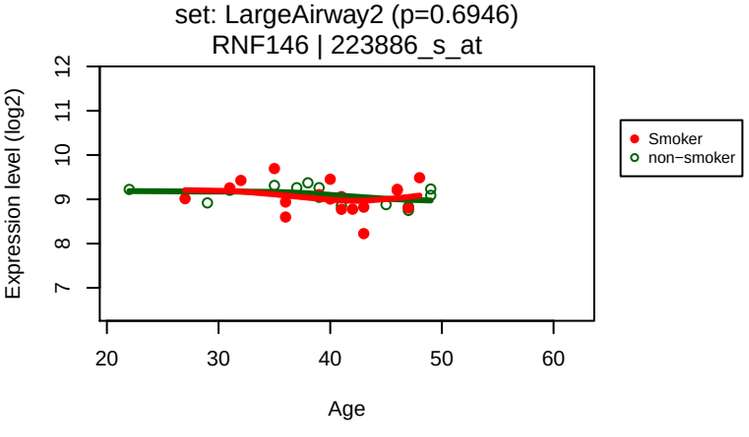

Gene expression ratio (smokers/nonsmokers)
gene: RNF146 | 223886_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3853
p=0.7887
p=0.4013
p=0.4083
p=0.6946
p=0.0638
p=0.4719
p=0.1479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

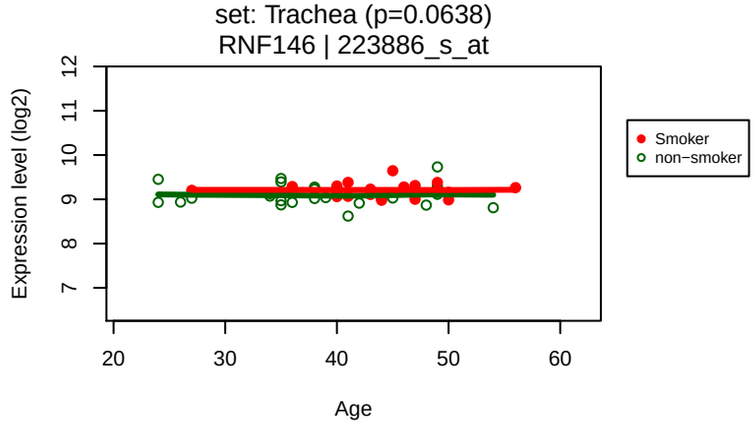

Gene expression ratio in Males
gene: RNF146 | 223886_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0804
p=0.1574
p=0.5313
p=0.1684
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNF146 | 223886_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3941
p=0.1802
p=0.0261
p=0.7125
SmallAirway2
LargeAirway2
Trachea
Macrophage2

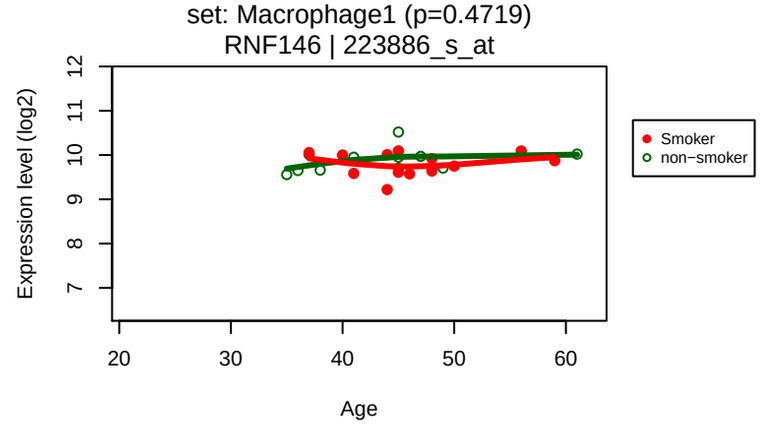

set: SmallAirway1 (p=0.3853)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker

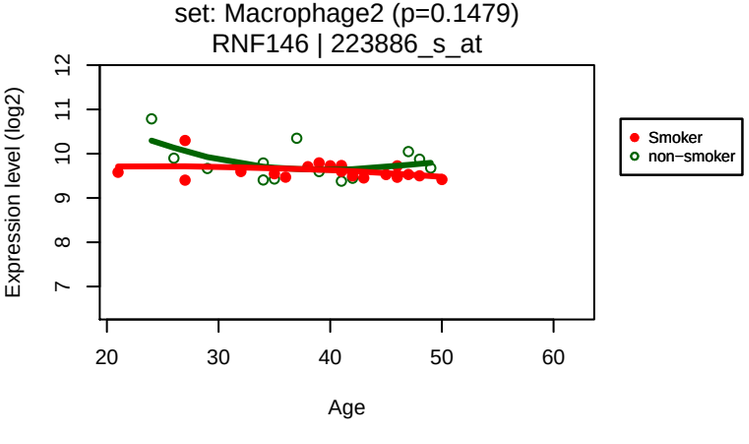

set: SmallAirway2 (p=0.7887)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker

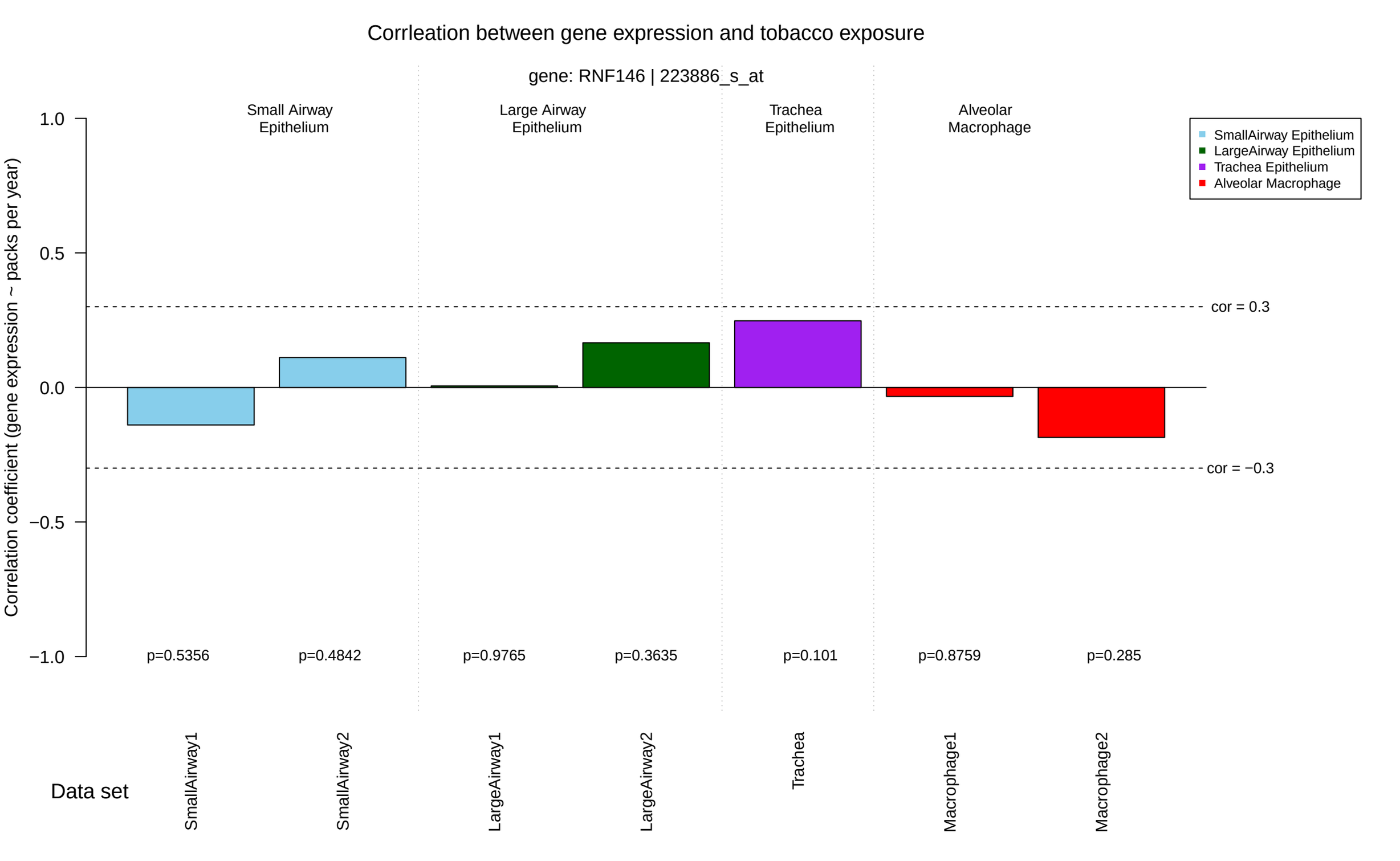

set: LargeAirway1 (p=0.4083)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6946)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0638)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4719)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1479)
RNF146 | 223886_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNF146 | 223886_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5356
p=0.4842
p=0.9765
p=0.3635
p=0.101
p=0.8759
p=0.285
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2